

Calculus

Star

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Chapter 13 Applications of Differential Calculus

1 Applications to Geometry

Definition 13.1.1 Suppose f is a scalar field on $S \subseteq \mathbb{R}^n$ and $c \in \mathbb{R}$ is a constant. We call the set $L(c) := \{\vec{x} \in S : f(\vec{x}) = c\}$ a **level set** of f . In particular, if $n = 2$, we call $L(c)$ a **level curve** of f ; if $n = 3$, we call $L(c)$ a **level surface** of f .

Example 13.1.2 Suppose $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. For any curve $\vec{r}(t)$ in $\mathbb{R}^2$, if we let g be defined as $g(t) = f[\vec{r}(t)]$, then

$$g'(t) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = \|\vec{r}'(t)\| (\nabla f(\vec{r}(t)) \cdot T(t)).$$

Particularly, if this curve happens to be the level curve given by $f(x, y) = c$. Suppose level curve $L(c) := \{(x, y) : f(x, y) = c\}$ is also described by $\vec{r}_c(t)$. Then $g(t) = f[\vec{r}_c(t)] = c$ for any t , which means $g'(t) = 0$ for any t . Thus

$$\|\vec{r}'_c(t)\| (\nabla f(\vec{r}_c(t)) \cdot T(t)) = 0.$$

On this level curve, $\nabla f \cdot T = 0$, and we say: the gradient vector is always normal to the curve C ; the directional derivative of f along any level curve of f is zero. By Cauchy-Schwarz inequality, we have

$$|\nabla f(x, y) \cdot \vec{y}| \leq \|\nabla f(x, y)\| \|\vec{y}\|,$$

maximized when $\vec{y}$ is parallel to $\nabla f(x, y)$, i.e., $\vec{y}$ is perpendicular to the tangent vector of a level curve.

Example 13.1.3 Suppose $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ differentiable. Suppose $L(c)$ is a level surface and $\vec{a}$ is a point on the level surface. Let $\vec{r}(t)$ describes a curve that passes through $\vec{a}$ and lies completely within $L(c)$. If we consider all curves passing through $\vec{a}$ and living within $L(c)$, then the tangent vectors of all these curves at $\vec{a}$ determine/live in a plane that passes through $\vec{a}$ and has a normal vector equal to $\nabla f(\vec{a})$ provided that $\nabla f(\vec{a}) \neq \vec{0}$. This plane is can be described by the equation

$$\nabla f(\vec{a}) \cdot (\vec{x} - \vec{a}) = 0.$$

2 Applications to Optimization Problems

Definition 13.2.1 A scalar field is said to

- have an **absolute (or global maximum)** at $\vec{a} \in S \subseteq \mathbb{R}^n$ if $f(\vec{a}) \geq f(\vec{x})$ for any $\vec{x} \in S$, in which case we also call $f(\vec{a})$ the **absolute maximum value** of f on S ;
- have an **absolute (or global minimum)** at $\vec{a} \in S$ if $f(\vec{a}) \leq f(\vec{x})$ for any $\vec{x} \in S$, in which case we also

call $f(\vec{a})$ the **absolute minimum value** of f on S ;

- have a **relative (or local maximum)** at $\vec{a} \in S$ if there exists $r > 0$ such that $f(\vec{a}) \geq f(\vec{x})$ for any $\vec{x} \in B(\vec{a}; r) \subseteq S$, in which case we also call $f(\vec{a})$ the **local maximum value** of f at $\vec{a}$;
- have a **relative (or local minimum)** at $\vec{a} \in S$ if there exists $r > 0$ such that $f(\vec{a}) \leq f(\vec{x})$ for any $\vec{x} \in B(\vec{a}; r) \subseteq S$, in which case we also call $f(\vec{a})$ the **local minimum value** of f at $\vec{a}$.

If f has either a relative maximum or a relative minimum at $\vec{a}$, then we say f has an **extremum** at $\vec{a}$.

Theorem 13.2.2 If f has an extremum at $\vec{a} \in \text{int } S$ and f is differentiable at $\vec{a}$, then $\nabla f(\vec{a}) = \vec{0}$.

Remark 13.2.3 If f is differentiable at $\vec{a}$ and $\nabla f(\vec{a}) = \vec{0}$, then f does not necessarily have an extremum at $\vec{a}$. Assume $z = f(x, y) = x^2 - y^2$, then $\nabla f(0, 0) = \vec{0}$ but f does not have an extremum at $(0, 0)$ since f can be positive or negative in any neighborhood of $(0, 0)$.

Definition 13.2.4 Suppose scalar field f is differentiable at $\vec{a}$. If $\nabla f(\vec{a}) = \vec{0}$, then we say $\vec{a}$ is a **stationary point** of f . A stationary point is called a **saddle point** if every open n -ball $B(\vec{a})$ contains both points $\vec{x}$ with $f(\vec{x}) > f(\vec{a})$ and points $\vec{y}$ with $f(\vec{y}) < f(\vec{a})$.

Remark 13.2.5 Suppose f has a stationary point at $\vec{a}$, then the nature of this stationary point is determined by the sign of $f(\vec{x}) - f(\vec{a})$ for $\vec{x}$ close to $\vec{a}$. Suppose $\vec{x} = \vec{a} + \vec{y}$, then

$$f(\vec{a} + \vec{y}) - f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{y} + \|\vec{y}\|E(\vec{a}, \vec{y}),$$

where $E(\vec{a}, \vec{y}) \rightarrow 0$ as $\vec{y} \rightarrow \vec{0}$.

Theorem 13.2.6 Suppose f is a scalar field with continuous second-order partial derivatives $D_{ij}f$ in an open n -ball $B(\vec{a})$. Then for all $\vec{y} \in \mathbb{R}^n$ such that $(\vec{a} + \vec{y}) \in B(\vec{a})$, we have

$$f(\vec{a} + \vec{y}) - f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{y} + \frac{1}{2!} \vec{y} H(\vec{a} + c\vec{y}) \vec{y}^T,$$

where $c \in (0, 1)$ and $H(\vec{x})$ is called the **Hessian matrix** whose ij -th entry is given by $D_{ij}f(\vec{x}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\vec{x})$. Note that here $\vec{y}$ is treated as a row vector. We can also write the second-order Taylor expansion as

$$f(\vec{a} + \vec{y}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot \vec{y} + \frac{1}{2!} \vec{y} H(\vec{a}) \vec{y}^T + \|\vec{y}\|^2 E_2(\vec{a}, \vec{y}),$$

where $E_2(\vec{a}, \vec{y}) \rightarrow 0$ as $\vec{y} \rightarrow \vec{0}$.

Example 13.2.7 Consider the function

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

By definition,

$$D_1 f(0, y) = \lim_{h \rightarrow 0} \frac{f(h, y) - f(0, y)}{h} = \lim_{h \rightarrow 0} y \frac{h^2 - y^2}{h^2 + y^2} = -y,$$

$$D_{2,1} f(0, 0) = D_2(D_1 f)(0, 0) = \lim_{k \rightarrow 0} \frac{D_1 f(0, k) - D_1 f(0, 0)}{k} = -1.$$

Analogously, we have

$$D_2 f(x, 0) = x, \quad D_{1,2} f(0, 0) = 1.$$

Though $D_{1,2} f(0, 0)$ and $D_{2,1} f(0, 0)$ both exist, they are not equal since we do not have the continuity of $D_{1,2} f$ and $D_{2,1} f$ at $(0, 0)$: note

$$D_{2,1} f(x, y) = \frac{x^6 + 9x^4 y^2 - 9x^2 y^4 - y^6}{(x^2 + y^2)^3}, \quad (x, y) \neq (0, 0),$$

and we have

$$D_{2,1} f(0, 0) = -1 \neq 1 = \frac{x^6}{x^6} = \lim_{x \rightarrow 0} D_{2,1} f(x, 0) \quad \text{along } y = 0.$$

This example shows that the continuity of second-order partial derivatives is necessary for the equality of mixed partial derivatives and motivates the following theorem.

Theorem 13.2.8 Suppose f is a scalar field on an open set $S \subseteq \mathbb{R}^n$, and suppose the first-order partial derivatives $D_i f$ and $D_j f$ exist on S . Assume further that the second-order partial derivatives $D_{ij} f$ and $D_{ji} f$ exist on S and are continuous at $\vec{a} \in S$. Then

$$D_{ij} f(\vec{a}) = D_{ji} f(\vec{a}).$$

Remark 13.2.9 One can actually prove a **stronger** version of the above theorem, which only requires the continuity of **one of the two** second order partial derivatives $D_{i,j} f$ and $D_{j,i} f$ at $\vec{a}$.

Lemma 13.2.10 By Theorem 13.2.8, we know the symmetric Hessian matrix has an orthonormal eigenbasis and $\vec{y} H(\vec{a} + c\vec{y}) \vec{y}^T$ is a quadratic form associated with a unique symmetric matrix. If we know all the eigenvalues of $H(\vec{a})$ are positive, then

$$f(\vec{a} + \vec{y}) - f(\vec{a}) = \frac{1}{2} \vec{y} H(\vec{a}) \vec{y}^T + \|\vec{y}\|^2 E_2(\vec{a}, \vec{y}) > 0$$

for $\vec{y}$ close enough to $\vec{0}$.

Proof. Fix $\vec{y}$ and define $g(u) = f(\vec{a} + u\vec{y})$ for $u \in [-1, 1]$. Then

$$f(\vec{a} + \vec{y}) = g(1) - g(0) = g'(0) + \frac{1}{2} g''(c)$$

where $c \in (0, 1)$ by the second-order Taylor formula for g on $[0, 1]$ with the Lagrange form of the remainder.

Let $\vec{r} = \vec{a} + u\vec{y}$, then $\vec{r}'(u) = \vec{y}$ and

$$g'(u) = \nabla f(\vec{r}) \cdot \vec{r}'(u) = \sum_{j=1}^n D_j f(\vec{r}) y_j.$$

Thus

$$g'(0) = \sum_{j=1}^{n=1} D_j f(\vec{a}) y_j = \nabla f(\vec{a}) \cdot \vec{y} = 0.$$

□

Lemma 13.2.11 Suppose $A \in \mathbf{M}_{n \times n}(\mathbb{R})$ is a symmetric matrix and $\vec{y}$ is a row vector in $\mathbb{R}^n$. Let $Q(\vec{y}) = \vec{y} A \vec{y}^T$. Then

1. $Q(\vec{y}) > 0$ for any $\vec{y} \neq \vec{0}$ if and only if all the eigenvalues of A are positive;
2. $Q(\vec{y}) < 0$ for any $\vec{y} \neq \vec{0}$ if and only if all the eigenvalues of A are negative;
3. $Q(\vec{y})$ takes both positive and negative values for $\vec{y} \neq \vec{0}$ if and only if A has both positive and negative eigenvalues.

Theorem 13.2.12 Suppose f is a scalar field with continuous second-order partial derivatives $D_{ij}f$ in an open n -ball $B(\vec{a})$ and Let $H(\vec{a})$ denote the Hessian matrix of f at $\vec{a}$, which is a stationary point of f . Then

1. If all the eigenvalues of $H(\vec{a})$ are positive, i.e., $H(\vec{a})$ is positive definite, then f has a local minimum at $\vec{a}$;
2. If all the eigenvalues of $H(\vec{a})$ are negative, i.e., $H(\vec{a})$ is negative definite, then f has a local maximum at $\vec{a}$;
3. If $H(\vec{a})$ has both positive and negative eigenvalues, then f has a saddle point at $\vec{a}$.

Corollary 13.2.13 Let $\vec{a}$ be a stationary point of a scalar field $f(x_1, x_2)$ with continuous second order partial derivatives in a 2-ball $B(\vec{a})$. Let

$$A = D_{1,1}f(\vec{a}), \quad B = D_{1,2}f(\vec{a}), \quad C = D_{2,2}f(\vec{a}),$$

and let

$$\Delta = \det H(\vec{a}) = \det \begin{pmatrix} A & B \\ B & C \end{pmatrix} = AC - B^2.$$

Then we have:

- (a) If $\Delta > 0$ and $A > 0$, then f has a local minimum at $\vec{a}$;
- (b) If $\Delta > 0$ and $A < 0$, then f has a local maximum at $\vec{a}$;
- (c) If $\Delta < 0$, then f has a saddle point at $\vec{a}$;
- (d) If $\Delta = 0$, then the test is inconclusive.

Remark 13.2.14 The determinant test is only applicable for $n = 2$. For $n \geq 3$, we need to check the signs of all the eigenvalues of the Hessian matrix to determine the nature of a stationary point.

Theorem 13.2.15 Given $\vec{a} = (a_1, \dots, a_n)$, $\vec{b} = (b_1, \dots, b_n) \in \mathbb{R}^n$ with $a_i < b_i$ for $i = 1, \dots, n$. We define $[\vec{a}, \vec{b}] := [a_1, b_1] \times \dots \times [a_n, b_n]$. If a scalar field f is continuous at every $\vec{x} \in [\vec{a}, \vec{b}]$, then

- f is bounded on $[\vec{a}, \vec{b}]$, i.e., there exists $M > 0$ such that $|f(\vec{x})| \leq M$ for any $\vec{x} \in [\vec{a}, \vec{b}]$;
- There exists $\vec{c}$ and $\vec{d}$ such that

$$f(\vec{c}) = \sup_{\vec{x} \in [\vec{a}, \vec{b}]} f(\vec{x}), \quad f(\vec{d}) = \inf_{\vec{x} \in [\vec{a}, \vec{b}]} f(\vec{x}).$$

Definition 13.2.16 Suppose $[\vec{a}, \vec{b}]$ and P_k is a partition of $[a_k, b_k]$ for $k = 1, \dots, n$. We call $P := P_1 \times \dots \times P_n$ a **partition** of $[\vec{a}, \vec{b}]$.

Definition 13.2.17 Let f be a continuous scalar field on $[\vec{a}, \vec{b}] \subseteq \mathbb{R}^n$ and let $M(f)$ and $m(f)$ denote the absolute maximum and minimum values of f on $[\vec{a}, \vec{b}]$, respectively. We call $M(f) - m(f)$ the **span** of f on $[\vec{a}, \vec{b}]$.

Theorem 13.2.18 Suppose f is a scalar field continuous on $[\vec{a}, \vec{b}]$. Then, for every $\varepsilon > 0$, there is a partition of $[\vec{a}, \vec{b}]$ such that the span of f on every subinterval of this partition is less than ε .

Chapter 14 Line Integrals

1 Basic Definitions and Properties

Definition 14.1.1 A function $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$ that is continuous on $[a, b]$ is also called a **path** in $\mathbb{R}^n$. If $\vec{r}'$ exists and is continuous on (a, b) , then we say $\vec{r}$ is a **smooth path**. If there is a partition of $[a, b]$ such that $\vec{r}$ is smooth on each subinterval of this partition, then we say $\vec{r}$ is a **piecewise smooth path**.

Definition 14.1.2 Let $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$ be a piecewise smooth path and $\vec{f}$ be a vector field defined and bounded on the image of $\vec{r}$. The line integral of $\vec{f}$ along $\vec{r}$ is defined as

$$\int_{\vec{r}} \vec{f} \cdot d\vec{r} := \int_a^b \vec{f}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

provided the limit on the right-hand side exists.

Remark 14.1.3 Suppose curve C is described by $\vec{r}(t)$ for $t \in [a, b]$ and $\vec{r}$ is a piecewise smooth path, then we can also write $\int \vec{f} \cdot d\vec{r}$ as

1. $\int_C \vec{f} \cdot d\vec{r}$;
2. $\int_{\vec{a}}^{\vec{b}} \vec{f} \cdot d\vec{r}$ where $\vec{a} = \vec{r}(a)$ and $\vec{b} = \vec{r}(b)$;
3. $\oint \vec{f} \cdot d\vec{r}$ or $\oint_C \vec{f} \cdot d\vec{r}$ if C is a closed curve, i.e., $\vec{r}(a) = \vec{r}(b)$;
4. $\int_C f_1(x, y)dx + f_2(x, y)dy$ if $\vec{f} = (f_1(x, y), f_2(x, y))$ with $x = r_1(t)$ and $y = r_2(t)$.

Remark 14.1.4 Since we define line integrals in terms of regular integrals of a scalar function on $[a, b]$, line integrals have the following basic properties of regular integral.

Theorem 14.1.5 Suppose $\vec{f}$ and $\vec{g}$ are vector fields defined and bounded on the image of a piecewise smooth path $\vec{r}$. Then

(1) Linearity:

$$\int (\alpha \vec{f} + \beta \vec{g}) \cdot d\vec{r} = \alpha \int \vec{f} \cdot d\vec{r} + \beta \int \vec{g} \cdot d\vec{r}$$

for any scalar α, β ;

(2) Additivity: if curve C can be regarded as two pieces C_1 and C_2 joined together, then

$$\int_C \vec{f} \cdot d\vec{r} = \int_{C_1} \vec{f} \cdot d\vec{r} + \int_{C_2} \vec{f} \cdot d\vec{r};$$

Definition 14.1.6 Suppose $\vec{\beta} : [a, b] \rightarrow \mathbb{R}^n$ is a continuous path and $u : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable on $[c, d]$ with $\{u(s) : s \in [c, d]\} = [a, b]$ and $u'(s) \neq 0$ for any $s \in [c, d]$. Then $\vec{\gamma} : [c, d] \rightarrow \mathbb{R}^n$ defined as $\vec{\gamma}(s) := \vec{\beta}[u(s)]$ for $s \in [c, d]$ is also a continuous path with the same image as $\vec{\beta}$. We call $\vec{\beta}$ and $\vec{\gamma}$ **equivalent paths** and u a **change of parameter function**. Namely, $\vec{\gamma}$ and $\vec{\beta}$ describe the same curve but with different parametrizations. If $u'(s) > 0$ for $s \in [c, d]$, then we say $\vec{\gamma}$ and $\vec{\beta}$ trace out C in the same direction.

Definition 14.1.7 Suppose $\vec{\alpha}$ is a path with $\vec{\alpha}'$ continuous on $[a, b]$. Then $\vec{\alpha}$ describes a **rectifiable curve** whose arc-length function is given by

$$s(t) = \int_a^t \|\vec{\alpha}'(\tau)\| d\tau, \quad t \in [a, b].$$

We also know $s'(t) = \|\vec{\alpha}'(t)\|$. Suppose f is a scalar function defined and bounded on C . The line integral of f with respect to arc-length along C is defined as

$$\int_C f ds := \int_a^b f(\vec{\alpha}(t)) s'(t) dt.$$

Example 14.1.8 Let $\vec{g}$ be a vector field (e.g. velocity field of the fluid) defined on a curve C described by $\vec{\alpha}(t)$ on $[a, b]$. Then $\varphi[\vec{\alpha}(t)] := \vec{g}[\vec{\alpha}(t)] \cdot T(t)$ is a scalar field defined on C and

$$\int_C \varphi ds = \int_a^b \vec{\varphi}[\vec{\alpha}(t)] s'(t) dt = \int_a^b \vec{g}[\vec{\alpha}(t)] \cdot T(t) s'(t) dt = \int_a^b \vec{g}[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t) dt = \int_C \vec{g} \cdot d\vec{r}.$$

Then $\int_C \varphi ds = \int_C \vec{g} \cdot T ds$ is called the **flow integral** of $\vec{g}$ along C . Particularly, when C is a closed curve, it is also called the **circulation** of $\vec{g}$ along C .

2 Fundamental Theorems of Calculus for Line Integrals

Definition 14.2.1 Let S be an open set in $\mathbb{R}^n$. We say S is **connected** if every pair of points in S can be joined by a piecewise smooth path whose image is totally contained in S .

Theorem 14.2.2 (Second Fundamental Theorem of Calculus for Line Integrals) Let f be a differentiable scalar field with continuous gradient ∇f on an open, connected set $S \subseteq \mathbb{R}^n$. Then for points $\vec{a}, \vec{b} \in S$ that are joined by a piecewise smooth path $\vec{\alpha}$ in S , we have

$$\int_{\vec{\alpha}} \nabla f \cdot d\vec{\alpha} = f(\vec{b}) - f(\vec{a}).$$

Proof. Let $\vec{a}, \vec{b} \in S$ and $\vec{\alpha}$ be a path with $\vec{\alpha}(a) = \vec{a}$ and $\vec{\alpha}(b) = \vec{b}$ and $\vec{\alpha}(t) \in S$ for $t \in [a, b]$.

Special Case: $\vec{\alpha}$ is smooth on $[a, b]$. We pick $g(t) := f[\vec{\alpha}(t)]$ for $t \in [a, b]$. Then g is differentiable on $[a, b]$ and

$$g'(t) = \nabla f[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t)$$

for $t \in [a, b]$. By the **FTC2** for regular integrals, we have

$$\int \nabla f \cdot d\vec{\alpha} = \int_a^b \nabla f[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t) dt = g(b) - g(a) = f(\vec{b}) - f(\vec{a}).$$

General Case: $\vec{\alpha}$ is piecewise smooth on $[a, b]$ with partition $P = \{t_0, t_1, \dots, t_r\}$ where $a = t_0 < t_1 < \dots < t_r = b$ such that $\vec{\alpha}$ is smooth on each subinterval $[t_{i-1}, t_i]$ for $i = 1, 2, \dots, r$. Then by the additive property

$$\int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \nabla f \cdot d\vec{\alpha} = \sum_{i=1}^r \int_{\vec{\alpha}(t_{i-1})}^{\vec{\alpha}(t_i)} \nabla f \cdot d\vec{\alpha} = \sum_{i=1}^r [f(\vec{\alpha}(t_i)) - f(\vec{\alpha}(t_{i-1}))] = f(\vec{b}) - f(\vec{a}).$$

□

Remark 14.2.3 The result of Theorem 14.2.2 can be interpreted as follows:

- the line integral's value of a gradient field only depends on the endpoints $\vec{a}$ and $\vec{b}$, i.e., **independent of path from $\vec{a}$ to $\vec{b}$** ;
- $\oint \nabla f \cdot d\vec{r} = 0$ for any closed curve in S .

Theorem 14.2.4 (First Fundamental Theorem of Calculus for Line Integrals) Let f be a vector field that is continuous on an open connected set $S \subseteq \mathbb{R}^n$ and assume that the line integral of f is independent of path in S . Let $\vec{a} \in S$ and define a scalar field φ on S by

$$\varphi(\vec{x}) := \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot d\vec{\alpha}, \quad \vec{x} \in S$$

where $\vec{\alpha}$ is any piecewise smooth path in S from $\vec{a}$ to $\vec{x}$. Then the gradient of φ exists and $\nabla\varphi(\vec{x}) = \vec{f}(\vec{x})$ on S .

Proof. Suppose $\vec{f}(\vec{x}) = (f_1(\vec{x}), f_2(\vec{x}), \dots, f_n(\vec{x}))$. We will prove $D_k\varphi(\vec{x}) = f_k(\vec{x})$ for $k = 1, 2, \dots, n$ and any $\vec{x} \in S$. Suppose $B(\vec{x}; r) \subseteq S$ and $\vec{y}$ is a unit vector. Then $\vec{x} + h\vec{y} \in B(\vec{x}; r)$ for any $0 < |h| < r$.

$$\frac{\varphi(\vec{x} + h\vec{y}) - \varphi(\vec{x})}{h} = \frac{1}{h} \left[\int_{\vec{a}}^{\vec{x} + h\vec{y}} \vec{f} \cdot d\vec{\alpha} - \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot d\vec{\alpha} \right] = \frac{1}{h} \int_{\vec{x}}^{\vec{x} + h\vec{y}} \vec{f} \cdot d\vec{\alpha}.$$

Particularly, consider the path $\vec{\alpha}(t) = \vec{x} + th\vec{y}$ for $t \in [0, 1]$. Then $\vec{\alpha}'(t) = h\vec{y}$ and

$$\frac{1}{h} \int_{\vec{x}}^{\vec{x} + h\vec{y}} \vec{f} \cdot d\vec{\alpha} = \int_0^1 \vec{f}(\vec{x} + th\vec{y}) \cdot \vec{y} dt = \int_0^1 f_k(\vec{x} + th\vec{e}_k) dt,$$

where $\vec{y} = \vec{e}_k$ is the k -th standard unit vector. We do a change of variable $u = th$ and get

$$\int_0^1 f_k(\vec{x} + th\vec{e}_k) dt = \frac{1}{h} \int_0^h f_k(\vec{x} + u\vec{e}_k) du = \frac{\int_0^h f_k(\vec{x} + u\vec{e}_k) du - \int_0^0 f_k(\vec{x} + u\vec{e}_k) du}{h - 0}$$

which converges to $f_k(\vec{x})$ as $h \rightarrow 0$ by the **FTC1** for regular integrals. Hence,

$$\lim_{h \rightarrow 0} \frac{\varphi(\vec{x} + h\vec{y}) - \varphi(\vec{x})}{h} = f_k(\vec{x}) = D_k \varphi(\vec{x}).$$

□

Remark 14.2.5 Suppose $\vec{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then by Theorem 14.2.4, if the line integral of $\vec{g}$ is independent of path in S , then there exists a scalar field φ such that $\nabla \varphi = \vec{g}$. Combining with Theorem 14.2.2, we have the necessary and sufficient condition for a vector field to be a gradient field, which is stated in the following theorem.

Theorem 14.2.6 Suppose $\vec{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then the following statements are equivalent:

- (1) $\vec{g}$ is the gradient of some scalar field φ in S , i.e., $\vec{g} = \nabla \varphi$, where φ is called a **potential function** of $\vec{g}$;
- (2) the line integral of $\vec{g}$ is independent of path in S ;
- (3) $\oint_C \vec{g} \cdot d\vec{r} = 0$ for any piecewise smooth closed path C in S .

Proof. By Theorem 14.2.4, we have (2) $\Rightarrow$ (1). By Theorem 14.2.2, we have (1) $\Rightarrow$ (2) and (1) $\Rightarrow$ (3).

We will only show (3) $\Rightarrow$ (2). Suppose C_1 and C_2 are any two piecewise smooth paths in S sharing the same endpoints. Let C_1 be described by $\vec{\alpha}$ on $[a, b]$ and C_2 be described by $\vec{\beta}$ on $[c, d]$ with $\vec{\alpha}(a) = \vec{\beta}(c)$ and $\vec{\alpha}(b) = \vec{\beta}(d)$.

Define $\vec{\gamma}$ as

$$\vec{\gamma}(t) = \begin{cases} \vec{\alpha}(t), & t \in [a, b]; \\ \vec{\beta}(b + d - t), & t \in [b, b + d - c]. \end{cases}$$

Note $\vec{\gamma}(a) = \vec{\alpha}(a) = \vec{\beta}(c) = \vec{\gamma}(b + d - c)$. Then $\vec{\gamma}$ is a piecewise smooth path describing a closed curve in S .

Then by (3), we know

$$0 = \int_{\vec{\gamma}(a)}^{\vec{\gamma}(b+d-c)} \vec{g} \cdot d\vec{r} = \int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot d\vec{r} + \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot (-d\vec{r}) = \int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot d\vec{r} - \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot d\vec{r}.$$

Hence, $\int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot d\vec{r} = \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot d\vec{r}$, which means the line integral of $\vec{g}$ is independent of path in S . □

Remark 14.2.7 Actually, Theorem 14.2.4 not only proves (2) $\Rightarrow$ (1), it also tells us we can find a potential function φ (such that $\nabla \varphi = \vec{f}$) by picking $\varphi(\vec{x}) := \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot d\vec{\alpha}$ for any $\vec{a} \in S$ and any piecewise smooth path $\vec{\alpha}$ from $\vec{a}$ to $\vec{x}$.

Example 14.2.8 We call $S \subseteq \mathbb{R}^n$ a **convex set** if every pair of points in S can be joined by a line segment fully contained in S . Note if S is open convex, then S is open connected. Suppose S is open convex, we can pick

$$\varphi(\vec{x}) := \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot d\vec{\alpha} = \int_0^1 \vec{f}[\vec{a} + t(\vec{x} - \vec{a})] \cdot (\vec{x} - \vec{a}) dt.$$

Particularly, if $\vec{0} \in S$, then $\varphi(\vec{x}) = \int_0^1 \vec{f}(t\vec{x}) \cdot \vec{x} dt$.

Example 14.2.9 Let $\vec{f} = (f_1, f_2, f_3)$, we know

$$\frac{\partial \varphi}{\partial x} = f_1, \quad \frac{\partial \varphi}{\partial y} = f_2, \quad \frac{\partial \varphi}{\partial z} = f_3.$$

Then we can pick

$$\varphi(x, y, z) = \int f_1(x, y, z) dx + C_1(y, z) = \int f_2(x, y, z) dy + C_2(x, z) = \int f_3(x, y, z) dz + C_3(x, y).$$

Note that the functions C_1, C_2, C_3 are not arbitrary constants, but functions of the remaining variables. In practice, we collect all terms from the three integrals $\int f_1(x, y, z) dx$, $\int f_2(x, y, z) dy$ and $\int f_3(x, y, z) dz$ and add the missing terms into $C_1(y, z)$, $C_2(x, z)$, or $C_3(x, y)$.

Chapter 15 Multiple Integrals

1 Double Integrals

In Chapter 02, we define the integral of a general function $f : [a, b] \rightarrow \mathbb{R}$ by approximating f by step functions. In this section, we follow the same idea to define the double integral of a function $f : Q \rightarrow \mathbb{R}$ where Q is a rectangle in $\mathbb{R}^2$.

Definition 15.1.1 Given two closed intervals $[a, b]$ and $[c, d]$ in $\mathbb{R}$, we call

$$Q := [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}$$

a **rectangle** in $\mathbb{R}^2$.

1. Suppose $P_1 = \{x_0, x_1, \dots, x_n\}$ and $P_2 = \{y_0, y_1, \dots, y_m\}$ are partitions of $[a, b]$ and $[c, d]$, respectively, with

$$a = x_0 < x_1 < \dots < x_n = b, \quad c = y_0 < y_1 < \dots < y_m = d.$$

Then we call

$$P := P_1 \times P_2 = \{[x_{i-1}, x_i] \times [y_{j-1}, y_j] : 1 \leq i \leq n, 1 \leq j \leq m\}$$

a **partition** of Q , which decomposes Q into nm **subrectangles**.

2. The Cartesian product $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$, where $1 \leq i \leq n$ and $1 \leq j \leq m$, is called an **open subrectangle** of P , or of Q .
3. Another partition $\tilde{P}$ is said to be **finer than** P , or a **refinement** of P , if $P \subseteq \tilde{P}$, i.e., all the points in P are also in $\tilde{P}$.
4. A scalar field $f : Q \rightarrow \mathbb{R}$ is called a **step function** if there exists a partition P of Q such that f is constant on each open subrectangle of P .

Remark 15.1.2 For a rectangle Q in $\mathbb{R}^2$, the set of all step functions defined on Q is a vector space over $\mathbb{R}$: if f and g are step functions defined on Q with respect to partitions P_f and P_g , respectively, then $\alpha f + \beta g$ is a step function defined on Q with respect to the common refinement $P_f \cup P_g$ of P_f and P_g for any $\alpha, \beta \in \mathbb{R}$.

Definition 15.1.3 Suppose $Q = [a, b] \times [c, d]$ with a partition $P = P_1 \times P_2$ where $P_1 = \{x_0, x_1, \dots, x_n\}$ and $P_2 = \{y_0, y_1, \dots, y_m\}$. Let f be a step function defined on Q that takes constant value c_{ij} on the open subinterval $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$ for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we define the **double integral** of f over Q by

$$\iint_Q f(x, y) dx dy := \sum_{i=1}^n \sum_{j=1}^m c_{ij} \cdot (x_i - x_{i-1})(y_j - y_{j-1}).$$

Note if we replace P with a finer partition $\tilde{P}$, then the value of the double integral does not change.

Remark 15.1.4 Since $f(x, y) = c_{ij}$ on $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$, if we let $Q_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$, then we have

$$\iint_{Q_{ij}} f(x, y) dx dy = c_{ij} \cdot (x_i - x_{i-1})(y_j - y_{j-1}) = f(x, y) \int_{x_{i-1}}^{x_i} dx \int_{y_{j-1}}^{y_j} dy = \int_{y_{j-1}}^{y_j} \left[\int_{x_{i-1}}^{x_i} f(x, y) dx \right] dy.$$